

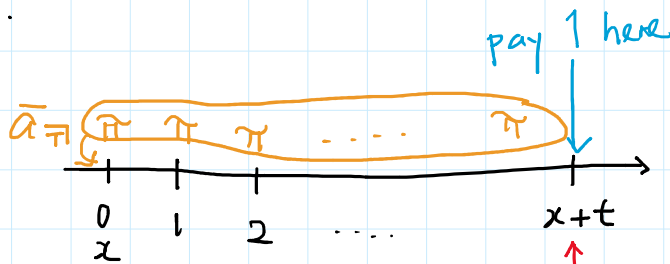
Loss on Endowment.

$$E({}_0L) = {}^1\bar{A}_{x:\overline{n}|} - \bar{P}({}^1\bar{A}_{x:\overline{n}|})$$

$\pi = \text{premium}$

$${}^1V({}_0L) = \left(1 + \frac{\bar{P}({}^1\bar{A}_{x:\overline{n}|})}{\delta}\right)^2 ({}^2\bar{A}_{x:\overline{n}|} - \bar{A}_{x:\overline{n}|}^2)$$

Proof.



Let's said you survived after T years
(and now you're $x+T$ years old).
What you really get is $PV(\text{Benefit})$
minus the annuity of premium
which you already paid.
Does this really payoff?

Death happens here

Let loss at issue be

$${}_0L = \begin{cases} {}^1V^T - \pi \bar{a}_{\overline{T}|} & 0 < T < n \\ {}^1V^n - \pi \bar{a}_{\overline{n}|} & T \geq n \end{cases} \quad \text{--- (1)}$$

$$= Z - \pi Y$$

where

$$Z = \begin{cases} V^T & 0 < T < n \\ V^n & T \geq n \end{cases} \Rightarrow E(Z) = {}^1\bar{A}_{x:\overline{n}|}$$

$$Y = \begin{cases} \bar{a}_{\overline{T}|} & 0 < T < n \\ \bar{a}_{\overline{n}|} & T \geq n \end{cases} \Rightarrow E(Y) = \bar{a}_{x:\overline{n}|}$$

Rewrite (1),

$$= \frac{1 - v^T}{1 - v} - \pi \frac{1 - v^T}{1 - v}$$

Return 

$${}_0L = \begin{cases} v^T - \pi \left(\frac{1-v^T}{\delta} \right) & 0 < T < n \\ v^n - \pi \left(\frac{1-v^n}{\delta} \right) & T \geq n \end{cases}$$

$$= Z - \frac{\pi}{\delta} + \frac{\pi}{\delta} Z$$

$$\Rightarrow {}_0L = \left(1 + \frac{\pi}{\delta} \right) Z - \frac{\pi}{\delta} \quad \text{---} \star$$

Taking Variance on $\star$,

$$V({}_0L) = V \left[\left(1 + \frac{\pi}{\delta} \right) Z \right] - \cancel{V \left(\frac{\pi}{\delta} \right)}^0$$

$$= \left(1 + \frac{\pi}{\delta} \right)^2 V(Z)$$

$$= \left(1 + \frac{\pi}{\delta} \right)^2 \left(E(Z^2) - E(Z)^2 \right)$$

$$= \left(1 + \frac{\pi}{\delta} \right)^2 \left[{}^2\bar{A}_{x:\overline{n}|} - (\bar{A}_{x:\overline{n}|})^2 \right] \quad \text{**}$$

Note: $\star$

$$\left(1 + \frac{\pi}{\delta} \right)^2 \left[{}^2\bar{A}_{x:\overline{n}|} - (\bar{A}_{x:\overline{n}|})^2 \right]$$

$\bar{P}(\bar{A}_{x:\overline{n}|})$

$\left. \begin{matrix} {}^2\bar{A}_{x:\overline{n}|} \\ \bar{A}_{x:\overline{n}|} \end{matrix} \right\} \text{Endowment}$

$\bar{P}(\bar{A}'_{x:\overline{n}|})$

$\left. \begin{matrix} {}^2\bar{A}'_{x:\overline{n}|} \\ \bar{A}'_{x:\overline{n}|} \end{matrix} \right\} \text{Term}$

$P(A_{x:n})$

$A_{x:n}, \bar{A}_{x:n} \}$ Term

$\bar{P}(\bar{A}_x)$

${}^2\bar{A}_x, \bar{A}_x \}$ Full Life.

$$(2) V({}_0L) = \frac{{}^2\bar{A}_{x:n} - (\bar{A}_{x:n})^2}{(\delta \bar{a}_{x:n})^2}$$

Proof.

Tips: $\bar{P}(\bar{A}_{x:n}) = \frac{\bar{A}_{x:n}}{\bar{a}_{x:n}}$

$\bar{a}_{x:n} = \frac{1 - \bar{A}_{x:n}}{\delta}$

$$\bar{P}(\bar{A}_{x:n}) = \frac{\delta \bar{A}_{x:n}}{1 - \bar{A}_{x:n}} = \pi$$

Substitute into **.

$$V({}_0L) = \left(1 + \frac{\pi}{\delta}\right)^2 [{}^2\bar{A}_{x:n} - (\bar{A}_{x:n})^2]$$

$$= \left(1 + \frac{1}{\cancel{\delta}} \left(\frac{\cancel{\delta} \bar{A}_{x:n}}{1 - \bar{A}_{x:n}} \right) \right)^2 [{}^2\bar{A}_{x:n} - (\bar{A}_{x:n})^2]$$

$$= \left(1 + \frac{\bar{A}_{x:n}}{1 - \bar{A}_{x:n}}\right)^2 [000]$$

$$= \left(\frac{1 - \cancel{\bar{A}_{x:n}} + \cancel{\bar{A}_{x:n}}}{1 - \bar{A}_{x:n}} \right)^2 [000]$$

$$= \left(\frac{1}{1 - \bar{A}_{x:n}} \right)^2 [000]$$

, π

$1 - \bar{A}_{x:n}$

$$\begin{aligned}
 &= \left(\frac{1}{1 - \bar{A}_{x:\overline{n}|}} \right)^{-} [0.00] \\
 &= \frac{[0.00]}{(\delta \bar{a}_{x:\overline{n}|})^2} \\
 &= \frac{{}^2\bar{A}_{x:\overline{n}|} - (\bar{A}_{x:\overline{n}|})^2}{(\delta \bar{a}_{x:\overline{n}|})^2} \quad \text{---}
 \end{aligned}$$

$\bar{a}_{x:\overline{n}|} = \frac{1 - \bar{A}_{x:\overline{n}|}}{\delta}$
 $\Rightarrow \delta \bar{a}_{x:\overline{n}|} = 1 - \bar{A}_{x:\overline{n}|}$

Eg. 5.2.2

① $A'_{70:\overline{5}|} = A_{70} - {}_5E_{70} A_{75} = 0.42818 - 0.73295(0.50868)$
 $= 0.05334$

$A_{70:\overline{5}|} = A'_{70:\overline{5}|} + {}_5E_{70} = 0.05334 + 0.73295$
 $= 0.78629$

$\ddot{a}_{70:\overline{5}|} = \frac{1 - A_{70:\overline{5}|}}{\frac{0.05}{1.05}} = \frac{1 - 0.78629}{\frac{0.05}{1.05}} = 4.48791$

$\therefore P'_{70:\overline{5}|} = \frac{A'_{70:\overline{5}|}}{\ddot{a}_{70:\overline{5}|}} = 0.12331$

②

${}_3E_{70} = \frac{l_{73}}{l_{70}} \times v^3 = \frac{87916.8}{91082.4} \times 1.05^{-3} = 0.842588$

$$\begin{aligned}\ddot{a}_{70:\overline{3}|} &= \ddot{a}_{70} - {}^3E_{70} \ddot{a}_{73} \\ &= 12.0083 - 0.842588(11.0081) \\ &= 2.733\end{aligned}$$

$$\therefore P_{70:\overline{3}|}^1 = \frac{{}^3E_{70}}{\ddot{a}_{70:\overline{3}|}} = 0.3083015$$

$$\textcircled{3} \quad {}^7E_{70} = \frac{l_{77}}{l_{70}} \times v^7 = \frac{81904.3}{91082.4} \times 1.05^{-7} = 0.63907$$

$$\begin{aligned}A'_{70:\overline{7}|} &= A_{70} - {}^7E_{70} A_{77} = 0.42818 - 0.63907(0.54217) \\ &= 0.081695\end{aligned}$$

$$\textcircled{5} P_{70:\overline{7}|}^1 = \frac{A'_{70:\overline{7}|}}{\ddot{a}_{70:\overline{7}|}} = \frac{0.081695}{4.48791} = 0.018203351 \quad \times$$

Thm 5.2 Proof.

$$\begin{aligned}\text{From RHS} &\Rightarrow \bar{P}(\bar{A}'_{x:\overline{h}|}) + \bar{P}(\bar{A}'_{x:\overline{h}|}) \bar{A}_{x+h} \\ &= \frac{\bar{A}'_{x:\overline{h}|} + ({}_hE_x(\bar{A}_{x+h}))}{\bar{a}_{x:\overline{h}|}} \\ &= \frac{\bar{A}'_{x:\overline{h}|} + {}_h|\bar{A}_x}{\bar{a}_{x:\overline{h}|}} \quad \text{Full Life.} \\ &= \frac{\bar{A}_x}{\bar{a}_{x:\overline{h}|}} = {}_1\bar{P}(\bar{A}_x) = \text{LHS}\end{aligned}$$

$$= \frac{A_x}{\bar{a}_{x:\overline{n}|}} = {}_h \bar{P}(\bar{A}_x) = \text{LHS} \quad \#$$

Ex. 5.2.3

x	l_x	A_x
35	100,000.00	0.151375
36	99,737.15	0.158245
37	99,455.91	0.165386
38	99,154.72	0.172804
39	98,831.91	0.180505
40	98,485.68	0.188492

$$\begin{aligned} \textcircled{1} \quad {}_5E_{35} &= A_{35:\overline{5}|}^1 = v^5 {}_5p_{35} = 1.06^{-5} \times \frac{l_{40}}{l_{35}} \\ &= \frac{98485.68}{100000 \times 1.06^5} = 0.7359423 \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad A_{35} &= A_{35:\overline{5}|}^1 + {}_5E_{35} (A_{40}) \\ \Rightarrow A_{35:\overline{5}|}^1 &= A_{35} - {}_5E_{35} A_{40} \\ &= 0.151375 - 0.7359423 (0.188492) \\ &= 0.0126558. \end{aligned}$$

$$A_{35:\overline{5}|} = A_{35:\overline{5}|}^1 + {}_5E_{35} = 0.7485981$$

$$\bar{a}_{35:\overline{5}|} = \frac{1 - A_{35:\overline{5}|}}{d} = \frac{1 - 0.7485981}{\left(\frac{0.06}{1+0.06}\right)} = 4.4414158$$

$$\bar{P}(\bar{A}_{35:\overline{5}|}^1) = \bar{A}_{35:\overline{5}|}^1 \quad ; \quad A_{35:\overline{5}|}^1$$

$$\begin{aligned}
 \bar{P}(\bar{A}'_{35:\overline{5}|}) &= \frac{\bar{A}'_{35:\overline{5}|}}{\ddot{a}'_{35:\overline{5}|}} = \frac{i}{\delta} \frac{A'_{35:\overline{5}|}}{\ddot{a}'_{35:\overline{5}|}} \\
 &= \frac{0.06}{\ln(1.06)} \times \frac{0.0126558}{4.4414158} \\
 &= 0.0029342
 \end{aligned}$$

✱

③

$$\begin{aligned}
 {}_5\bar{P}(\bar{A}_{35}) &= \frac{i}{\delta} \left[P(A'_{35:\overline{5}|}) + P(A_{35:\overline{5}|}) A_{40} \right] \\
 &= \frac{0.06}{\ln(1.06)} \left[\frac{A'_{35:\overline{5}|} + {}_5E_{35} A_{40}}{\ddot{a}'_{35:\overline{5}|}} \right] \\
 &= \frac{0.06}{\ln(1.06)} \left[\frac{0.0126558 + 0.7359723(0.188492)}{4.4414158} \right] \\
 &= 0.0350965
 \end{aligned}$$

✱

Eg. 5.3.1

Let X_i = PV of monthly annuity for $(x)_i$ $\forall i=1, 2, \dots, 200$

$S = X_1 + X_2 + \dots + X_{200}$ = PV of aggregated payments.

$i = 0.06$, Convertible monthly,

$i(12)$, 12

convertible monthly

$$\left(1 - \frac{d^{(12)}}{12}\right)^{12} = 1 - \frac{0.06}{1.06}$$

$$\Rightarrow d^{(12)} = 12 \left[1 - \left(1 - 0.056604\right)^{1/12} \right] = 0.05813$$

$$E(X_i) = \ddot{a}_{62}^{(12)} = \frac{1 - A_{62}^{(12)}}{d^{(12)}} = \frac{1 - 0.4075}{0.05813} = 10.19267$$

$$V(X_i) = \frac{{}^2A_{62}^{(12)} - (A_{62}^{(12)})^2}{(d^{(12)})^2} = \frac{0.2105 - 0.4075^2}{0.05813^2} = 13.15255$$

By CLT, $E(S) = 200 \times \overbrace{180}^{90\% \text{ among } 200 \text{ life}} \times 10.19267 = 366936.12$

$$V(S) = 200 \times 180^2 \times 13.15255 = 85228.524$$

$$\Pr(S \leq 200\pi) = 0.9$$

90th percentile,

$$200\pi = E(S) + \overset{1.282}{Z_{0.9}} \sqrt{V(S)}$$

$$200\pi = 378771.45$$

$$\pi = 1893.86 \quad \#$$

Equivalence principle *

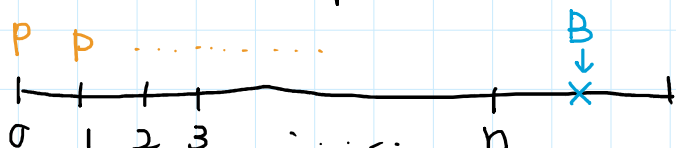
EPV of benefits = EPV of premiums.

EPV of benefits = EPV of premiums.

Proof.

Consider a whole life insurance policy of B is paid at the end of death.

Level annual premium P are paid.



$$\text{EPV of benefits} = BA_x$$

$$\text{EPV of premiums} = P \ddot{a}_x$$

$$\text{EPV of expenses} = E$$

Express P in term of A_x , the death benefit for (x) .

$$P \ddot{a}_x = BA_x + E$$

$$P = \frac{BA_x + E}{\ddot{a}_x} = \frac{BA_x + E}{\left[\frac{1 - A_x}{d} \right]}$$

$$= B \left[\frac{dA_x}{1 - A_x} \right] + \frac{Ed}{1 - A_x}$$

$$P = \frac{d(BA_x + E)}{1 - A_x} \quad *$$

Eg. 5.4.2

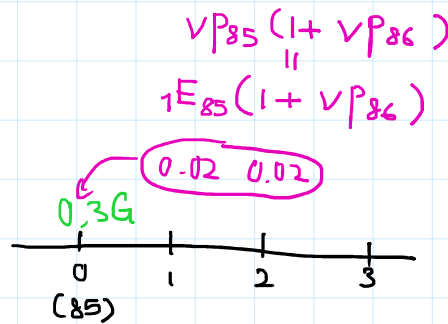
$$E(in) = G \ddot{a}_{55:31}$$

Benefits

$$E(in) = G \ddot{a}_{85:\overline{3}|}$$

$$E(out) = 100\,000 A'_{85:\overline{3}|} + G (IA)'_{85:\overline{3}|} + 500 + 0.3G + 0.02G (v p_{85} + v^2 p_{85} p_{86})$$

Expenses



$$E(in) = E(out)$$

$$G \ddot{a}_{85:\overline{3}|} = 100\,000 A'_{85:\overline{3}|} + G (IA)'_{85:\overline{3}|} + 500 + 0.3G + 0.02G (v p_{85} + v^2 p_{85} p_{86})$$

$$G(2.1900826 - 1.083396 - 0.3 - 0.02 \times 1.19) = 54845.98047 + 500$$

$$G = \frac{55345.98047}{0.7828866} = 70694.76022$$

$$\begin{aligned} \ddot{a}_{85:\overline{3}|} &= 1 + v p_{85} + v^2 {}_2p_{85} \\ &= 1 + \frac{0.8}{1.10} + \frac{0.8 \times 0.7}{1.1^2} = 2.1900826 \end{aligned}$$

$$\begin{aligned} (IA)'_{85:\overline{3}|} &= \sum_{k=0}^2 (k+1) v^{k+1} {}_k| \ddot{q}_{85} \\ &= \sum_{k=0}^2 (k+1) v^{k+1} {}_k p_{85} \ddot{q}_{85+k} \\ &= v \ddot{q}_{85} + 2v^2 p_{85} \ddot{q}_{86} + 3v^3 \underbrace{{}_2 p_{85}}_{p_{85} \times p_{86}} \ddot{q}_{87} \end{aligned}$$

$$= \frac{0.2}{1.1} + 2 \left(\frac{0.8 \times 0.3}{1.1^2} \right) + 3 \left(\frac{0.8 \times 0.7 \times 0.4}{1.1^3} \right)$$

$$= 1.083396$$

$$A'_{85:\overline{3}|} = \sum_{k=0}^2 v^{k+1} {}_k|q_{85} = \sum_{k=0}^2 v^{k+1} {}_k p_{85} q_{85+k}$$

$$= \frac{0.2}{1.1} + \frac{0.8 \times 0.3}{1.1^2} + \frac{0.8 \times 0.7 \times 0.4}{1.1^3}$$

$$= 0.5484598$$

$$\Rightarrow 100\,000 A'_{85:\overline{3}|} = 54845.98047$$

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